Joseph Simeon

Embedded software engineer, Brisbane, QLD.

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Embedded software engineer with over 4 years of expertise in C/C++ application and drivers in both bare-metal and RTOS for developing, testing, and supporting a smart touch LCD for lighting control on STM32 hardware using ST's TouchGFX. Strong background in electronics with a focusing on firmware and software programming, debugging and release testing a product that was used on a DALI cloud network.

Skills

- Programming in C/C++ for embedded systems (bare-metal, FreeRTOS) with knowledge in Unix, Python, Rust, HTML/CSS/JS, and SQL.
- Industry experience in application and driver programming for peripherals that interface with UART, I2C, SPI, DSI, DMA, QSPI with experience with EEPROM, LCDs, PWM, Timers, ADC, DMA.
- Control experience with docker, git, Jenkins utilising ticketing and tracking with Gitlab, and Redmine.
- Writing documentation in issue tracking completion, FATs & SATs, project development, release testing & tracking, and manuals for use.
- Creating embedded UX with custom widget/events within the graphical framework's scope.
- Providing customer and integrator support related to the product.

Experience

Firmware Engineer

Zencontrol, Brisbane — October 2020 — December 2024

- Handled development for smart touch LCD project, creating UI/UX with custom widgets and events functionality using graphics framework that interacted with a DALI controlled lighting system.
- Programming, debugging and release testing firmware built in C and C++ on STM32 hardware.
  - Testing the product in factory and site settings.
- Documentation writing on project development, wikipedia style manuals, and testing tickets.
- Providing custom support, collaborating with clients in troubleshooting issues to be turned into fixes.

Firmware Engineer (Thesis)

Griffith University, Brisbane — January 2020 — July 2020

- Creating a digital instrument cluster using Raspberry Pi and LCD with a graphical interface using Java and JavaFX.
- Up-skilling in all areas needed to complete the project such as learning a new language and framework.
- Independent research, utilising information to prove design decisions.
- Reading documentation such as data-sheets, standards, and equipment manuals.

Education

Bachelor of Engineering with Honours in Electronics and Energy Engineering

Griffith University — 2016 — 2021

- Assisted in creation of the SCADA course for Griffith University with other selected students.
- Membership of the IEEE society at Griffith University as a 2nd year — 4th year student, assisted in open events and member activities within Griffith University campus.

Diploma of Engineering

Griffith College (formerly QIBT) — 2014 — 2015

References can be provided on request